Somadutta Bhatta

Postdoctoral Researcher, High-Energy Nuclear Physics
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Research interests

Event-by-event fluctuations in heavy-ion collisions, studied through experimental measurements and phenomenological studies, to understand and constrain the quark-gluon plasma: radial and anisotropic flow, transverse-momentum fluctuations, the initial state of heavy-ion collisions, and nuclear deformation.

Positions

Postdoctoral Researcher 07/2025 – present
Institute for Gravitational and Subatomic Physics (GRASP), Utrecht University, and Nikhef, The Netherlands. ALICE Collaboration.
Supervisor: Prof. dr. R. J. M. (Raimond) Snellings

Education

Ph.D. in Physics 08/2018 – 06/2025
Stony Brook University (SUNY), New York, USA. Advisor: Prof. Jiangyong Jia.
Thesis: *Novel Probes of Quark-Gluon Plasma Evolution in Heavy-Ion Collisions With the ATLAS Detector: From Structure of Colliding Nuclei to Collective Expansion of Medium*, arXiv:2508.13764

Integrated B.Sc. and M.Sc. in Physics 08/2013 – 07/2018
National Institute of Science Education and Research (NISER), HBNI, Jatni, India. Advisor: Prof. Bedangadas Mohanty.
Thesis: *Understanding the mechanism of particle production in heavy-ion collisions at* $\sqrt{s_{\mathrm{NN}}} = 200$ *GeV*

Publications

Full lists: INSPIRE-HEP, Google Scholar, arXiv. Collaboration papers list authors alphabetically; my role is stated for each.

First/primary-author experimental papers

1. ATLAS Collaboration, *Evidence for the collective nature of radial flow in Pb+Pb collisions with the ATLAS detector*, Phys. Rev. Lett. **136**, 032301 (2026), Editors' Suggestion, arXiv:2503.24125.
   My role: lead analyzer; internal analysis note; paper writing.
   First measurement of the p_{T} dependence of radial-flow fluctuations, $v_0(p_{\mathrm{T}})$. It shows long-range correlation in pseudorapidity, factorization in p_{T} and a centrality-independent shape, the hallmarks of collectivity; a hydrodynamic comparison shows its sensitivity to bulk viscosity.
2. STAR Collaboration, *Imaging shapes of atomic nuclei in high-energy nuclear collisions*, Nature **635**, 67–72 (2024), arXiv:2401.06625.
   My role: Trajectum simulations for the extraction of the deformation parameters; paper writing.
   Introduced collective-flow-assisted nuclear shape imaging. From the comparison of U+U and Au+Au collisions, the quadrupole deformation and triaxiality of ^{238}U were extracted, consistent with low-energy experiments.
3. ATLAS Collaboration, *Disentangling sources of momentum fluctuations in Xe+Xe and Pb+Pb collisions with the ATLAS detector*, Phys. Rev. Lett. **133**, 252301 (2024), arXiv:2407.06413.

My role: lead analyzer; internal analysis note; paper writing.
Measurement of the mean, variance and skewness of the event-by-event $[p_{\mathrm{T}}]$ distribution. Their behaviour in ultra-central collisions, where geometrical fluctuations are suppressed, separates geometrical from intrinsic fluctuations and constrains the speed of sound of the quark-gluon plasma.

4. ATLAS Collaboration, *Correlations between flow and transverse momentum in Xe+Xe and Pb+Pb collisions at the LHC with the ATLAS detector: a probe of the heavy-ion initial state and nuclear deformation*, Phys. Rev. C **107**, 054910 (2023), arXiv:2205.00039.
My role: lead analyzer; internal analysis note; paper writing.
Measurement of the correlation between v_n^2 and $[p_{\mathrm{T}}]$. First experimental measurement showing that collective observables are sensitive to higher-order nuclear deformation and can be used to constrain it; the comparison of Xe+Xe and Pb+Pb gives strong evidence that ^{129}Xe is triaxial.

First/primary-author phenomenological papers

5. **S. Bhatta**, A. Dimri, J. Jia, *The shape of differential radial flow $v_0(p_{\mathrm{T}})$, not its zero-crossing, carries physical information*, Nucl. Sci. Tech. **37**, 163 (2026), arXiv:2504.20008.
6. V. Bairathi, **S. Bhatta**, *Method to constrain preferential emission and spectator dynamics in heavy-ion collisions*, Phys. Rev. C **113**, 024917 (2026), arXiv:2407.06977.
My role: corresponding author.
7. **S. Bhatta**, C. Zhang, J. Jia, *Energy dependence of heavy-ion initial condition in isobar collisions*, Phys. Lett. B **858**, 139034 (2024), arXiv:2301.01294.
8. **S. Bhatta**, C. Zhang, J. Jia, *Higher-order transverse momentum fluctuations in heavy-ion collisions*, Phys. Rev. C **105**, 024904 (2022), arXiv:2112.03397.
9. **S. Bhatta**, V. Bairathi, *An improved method to access initial states in relativistic heavy-ion collisions*, Eur. Phys. J. C **82**, 855 (2022), arXiv:2208.05157.

Other papers as contributing author

10. J. Jia, **S. Bhatta**, S. Huang, Z. Tu, C. Zhang, *Imaging Nuclei by Smashing Them: A New Way to Unveil the Nuclear Shapes*, Nucl. Phys. News **35**(1), 13–17 (2025).
11. J. Jia, S. Huang, C. Zhang, **S. Bhatta**, *Sources of longitudinal flow decorrelations in high-energy nuclear collisions*, arXiv:2408.15006 (2024), submitted.
12. R. Samanta, **S. Bhatta**, J. Jia, M. Luzum, J.-Y. Ollitrault, *Thermalization at the femtoscale seen in high-energy Pb+Pb collisions*, Phys. Rev. C **109**, L051902 (2024), arXiv:2303.15323.
13. A. Dimri, **S. Bhatta**, J. Jia, *Impact of nuclear shape fluctuations in high-energy heavy ion collisions*, Eur. Phys. J. A **59**, 45 (2023), arXiv:2301.03556.
14. C. Zhang, **S. Bhatta**, J. Jia, *Ratios of collective flow observables in high-energy isobar collisions are insensitive to final-state interactions*, Phys. Rev. C **106**, L031901 (2022), arXiv:2206.01943.
15. C. Zhang, A. Behera, **S. Bhatta**, J. Jia, *Non-flow effects in correlation between harmonic flow and transverse momentum in nuclear collisions*, Phys. Lett. B **822**, 136702 (2021), arXiv:2102.05200.

Proceedings and conference notes

16. V. Bairathi, **S. Bhatta**, *Deciphering initial states of high energy heavy-ion collisions using spectators*, Int. J. Mod. Phys. E **35**, 2642004 (2026).
17. **S. Bhatta** (for the ATLAS Collaboration), *Measurement of collective dynamics in small and large systems with the ATLAS detector*, PoS EPS-HEP2023, 202 (2024).
18. ATLAS Collaboration, *Measurement of transverse momentum fluctuations in Xe+Xe and Pb+Pb collisions with ATLAS*, ATLAS-CONF-2023-061 (2023).
My role: lead analyzer.
19. **S. Bhatta** (for the ATLAS Collaboration), *Flow and transverse momentum correlation in Pb+Pb and Xe+Xe collisions with ATLAS: assessing the initial condition of the QGP*, Acta Phys. Pol. B Proc. Suppl. **16**, 1-A90 (2023), arXiv:2210.00159.
20. **S. Bhatta** (for the ATLAS Collaboration), *Measurements of collective behavior in pp, Xe+Xe,*

and Pb+Pb collisions with the ATLAS detector, PoS EPS-HEP2021, 305 (2022).

21. ATLAS Collaboration, *Measurement of flow and transverse momentum correlations in Pb+Pb collisions at* $\sqrt{s_{\mathrm{NN}}} = 5.02$ *TeV and Xe+Xe collisions at* $\sqrt{s_{\mathrm{NN}}} = 5.44$ *TeV with the ATLAS detector*, ATLAS-CONF-2021-001 (2021).
    My role: lead analyzer.

Internal collaboration notes (primary author)

- ATLAS: Transverse dynamics of mean transverse momentum fluctuations in Pb+Pb collisions at $\sqrt{s_{\mathrm{NN}}} = 5.02$ TeV.
- ATLAS: Measurement of the mean transverse momentum fluctuations of charged particles in Xe+Xe collisions at $\sqrt{s_{\mathrm{NN}}} = 5.44$ TeV and Pb+Pb collisions at $\sqrt{s_{\mathrm{NN}}} = 5.02$ TeV.
- ATLAS: Measurement of flow and transverse momentum correlations in Xe+Xe collisions at $\sqrt{s_{\mathrm{NN}}} = 5.44$ TeV and Pb+Pb collisions at $\sqrt{s_{\mathrm{NN}}} = 5.02$ TeV.
- STAR: Imaging shapes of atomic nuclei in high-energy nuclear collisions.
- STAR: Beam-energy dependence of transverse momentum–flow correlations in Au+Au collisions.

Talks and posters

Invited talks and seminars

- CERN-TH Seminar (2025): Insights into Nuclear Geometry and Initial Conditions of Heavy-Ion Collisions from ATLAS.
- HENPIC Seminar (2024): Insights into Nuclear Geometry and Initial Conditions of Heavy-Ion Collisions from ATLAS.
- SBU HEP Seminar (2024): Leveraging Ultra-Central Collisions to Constrain Shape of the Nucleus and Initial State.
- Nuclear Physics Seminars at BNL (2024): Disentangling sources of momentum fluctuations in heavy-ion collisions with the ATLAS detector.
- LHCP (2024): Probing medium properties in ultra-central collisions (parallel talk on behalf of ATLAS+ALICE+CMS).
- INT Program 2023, Intersection of nuclear structure and high-energy nuclear collisions: Energy dependence of initial condition from isobar.
- Nuclear Physics Seminars at BNL (2022): Flow and transverse momentum correlations at LHC: a probe of the heavy-ion initial state and nuclear deformation.

International conference talks

- Initial Stages (2025): Recent flow and correlation measurements in large and small collision systems with ATLAS.
- Quark Matter (2025): Evidence for the collective nature of radial flow in Pb+Pb collisions with the ATLAS detector.
- ATHIC (2025): Unveiling initial state fluctuations using $[p_{\mathrm{T}}]$ cumulants with ATLAS.
- ICNFP (2024): Investigation of sources affecting longitudinal particle production in heavy-ion collisions using spectators.
- EPS-HEP (2023): Measurement of collective dynamics in small and large systems with the ATLAS detector.
- Initial Stages (2023): Probing initial state using higher order fluctuations: v_n–$[p_{\mathrm{T}}]$ and $[p_{\mathrm{T}}]$ correlations in ATLAS.
- Quark Matter (2022): Flow and transverse momentum correlation in Pb+Pb and Xe+Xe collisions with ATLAS.
- EPS-HEP (2021): Measurements of collective behavior in pp, Xe+Xe, and Pb+Pb collisions with the ATLAS detector.
- ATLAS TDAQ Week at CERN (2020): MinBias and Forward Detector.

National conference talks

- Fall Meeting of the APS Division of Nuclear Physics (2022): Probing nuclear deformation at LHC energies using AMPT.
- APS April Meeting (2022): Higher order transverse momentum fluctuations in heavy ion collisions.
- Fall Meeting of the APS Division of Nuclear Physics (2021): Flow and transverse momentum correlations in Pb+Pb and Xe+Xe collisions with ATLAS.
- APS April Meeting (2021): Cumulant analysis of deformed systems using AMPT model.

Posters

- Quark Matter (2023): Exploring the origin of $[p_{\mathrm{T}}]$ fluctuations in ultra-central heavy ion collisions.
- SQM (2022): Flow and transverse momentum correlations in Pb+Pb and Xe+Xe collisions with ATLAS.
- Quark Matter (2022): Higher order transverse momentum fluctuations in heavy-ion collisions.
- Initial Stages (2021): Measurement of flow and transverse momentum correlations in Pb+Pb and Xe+Xe collisions.
- APS Fall Meeting (2020): Flow cumulants for multi-particle azimuthal correlations in heavy-ion Collisions.

Awards and fellowships

- Distinguished Travel Award, Stony Brook University 2025
- Young Scientist Award, Initial Stages Conference 2023
- CSIR-NET (Physics) Junior Research Fellowship, All India Rank 39 2018
- Best Project Student, Department of High Energy Physics, TIFR, India 2016
- NIUS Scholar, Homi Bhabha Centre for Science Education, TIFR, India 2014–2015
- INSPIRE Fellowship, Department of Science & Technology, Government of India 2013–2018
- NTSE Scholarship, NCERT, Government of India 2009–2013

Supervision and teaching

- Master's thesis supervision, Utrecht University: Anne Dijkstra, *Mean transverse momentum fluctuations for strange and non-strange baryons in ALICE.*
- Bachelor's thesis supervision (completed), Utrecht University: Wouter Lansdaal, *Studying the rapidity dependence of mean transverse momentum using Trajectum.*
- Graduate Teaching Assistant, Stony Brook University (Spring 2018, Fall 2018, Spring 2019): undergraduate physics laboratory courses for more than 60 students.

Service

- Referee for Physical Review C, Physical Review D, Physics Letters B, Journal of Subatomic Particles and Cosmology, and Acta Physica Polonica B.
- Organizing Committee, XXVIII International Workshop on Deep-Inelastic Scattering (DIS 2021), Stony Brook University.
- ATLAS: development and maintenance of the tracking-efficiency and fake-rate tool for the heavy-ion group (2023–2024); migration of the minimum-bias, MBTS and high-multiplicity triggers to the multi-threaded Athena framework (2019–2022).
- STAR: detector-operator shifts (2019–2023).

Collaborations and skills

- Collaborations: ALICE (member); ATLAS and STAR (Ph.D. research).
- Programming and analysis: C++, Python, ROOT, Mathematica.

- Models: initial-state (Glauber, TRENTo), string-based (HIJING, AMPT) and hydrodynamic (Trajectum) frameworks, used in published work.

Media coverage

- World's most powerful particle collider reveals nature of quark soup's radial expansion, Interesting Engineering, 24 January 2026.
- ATLAS confirms collective nature of quark soup's radial expansion, BNL Newsroom, 22 January 2026; also Stony Brook University News, Newswise and ScienceSprings.
- A new probe of radial flow, CERN Courier, 8 July 2025.
- Scientists observe collective behavior of femtoscopic droplets at CERN, Phys.org, 16 May 2025.
- Stony Brook physicists unveil groundbreaking method for nuclear imaging, The Statesman, 22 November 2024.

Schools and training

- Exploring Nuclear Physics across Energy Scales, China Center of Advanced Science and Technology, Beijing 04/2024
- INT Program 23-1a, Intersection of nuclear structure and high-energy nuclear collisions, University of Washington 02/2023
- National Nuclear Physics Summer School, MIT 07/2022
- ATLAS HLT Upgrade Hackathon, CERN 06/2019

Earlier research experience

- Summer research, Saha Institute of Nuclear Physics, Kolkata: Garfield++ simulations of RPC and GEM detectors 05–07/2017
- Visiting Students' Research Programme, TIFR, Mumbai: multivariate (TMVA) selection of single-top events in CMS data 05–07/2016
- Semester project, NISER: light-nuclei production and elliptic flow in AMPT Au+Au collisions at 200 GeV 01–05/2016
- NIUS Physics Programme, HBCSE, TIFR, Mumbai: neutrino oscillations in vacuum and matter 2013–2015